Stage 2 report

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1 Literature review

1.1 Random-coefficients demand estimation and BLP

Empirical demand estimation for differentiated products is commonly conducted within the random-coefficients discrete-choice framework, in which consumers choose among products in a market based on observed characteristics and unobserved product–market shocks. The canonical approach in this literature is the Berry–Levinsohn–Pakes (BLP) model, which combines a random-utility specification with aggregate market share data and addresses price endogeneity through instrumental variables (Berry, 1994; Berry et al., 1995). Estimation proceeds via a contraction mapping that inverts observed shares to recover mean utilities, followed by GMM estimation of linear parameters and nonlinear search over random-coefficient parameters.

A central challenge in BLP-type models is the identification and estimation of unobserved demand shocks, typically denoted ξ_{jt}, that are correlated with endogenous prices. The performance of the estimator depends critically on the strength and validity of instruments, such as cost shifters or supply-side variables. When instruments are weak or unavailable, BLP estimators can exhibit substantial bias and dispersion, motivating alternative approaches that impose additional structure on unobserved shocks or exploit regularization to stabilize inference.

1.2 Sparse unobserved shocks and shrinkage in demand models

An emerging line of work introduces structure on unobserved demand shocks, particularly sparsity, whereby only a subset of product–market pairs experience economically meaningful idiosyncratic shocks. In such models, unobserved demand components are decomposed into a latent inclusion indicator and a shock magnitude, closely related to spike-and-slab priors and other shrinkage formulations used in high-dimensional Bayesian inference. By shrinking small or noisy shocks toward zero, these approaches can improve estimation accuracy and mitigate the impact of weak identification.

Lu and Shimizu (2025) adopt this perspective by modeling product–market demand shocks as sparse and by developing estimators that explicitly exploit this structure. Their framework allows the researcher to recover both the magnitude and the location of nonzero shocks while maintaining consistency in the presence of endogeneity. Through simulation designs with exogenous and endogenous sparse shocks, they demonstrate that shrinkage-based estimators can substantially reduce bias and variance relative to conventional BLP estimators, especially in settings where cost instruments are weak or omitted.

1.3 Bayesian computation: from Metropolis–Hastings to TMCMC and HMC

Estimating demand models with random coefficients and latent sparsity structures requires computational methods capable of exploring high-dimensional posterior distributions with strong dependence across parameters. Classical Metropolis–Hastings algorithms provide a general framework but often suffer from poor mixing in such settings. To address this, a variety of locally adapted or tailored MCMC methods have been proposed, including proposals based on local curvature or Laplace approximations. These methods, sometimes referred to as tailored or transformed MCMC (TMCMC), use information from the likelihood to construct more efficient proposal distributions and have been applied in structural IO contexts.

More recently, gradient-based samplers such as Hamiltonian Monte Carlo (HMC) and its adaptive variant, the No-U-Turn Sampler (NUTS), have become standard tools for sampling from complex continuous posteriors (Neal, 2011; Hoffman and Gelman, 2014). By leveraging gradient information, HMC can generate distant proposals with high acceptance probability, improving scalability in moderately high-dimensional parameter spaces. However, HMC does not directly accommodate discrete latent variables, such as inclusion indicators in sparse-shock models. A common solution is to adopt a blocked Gibbs sampling strategy, in which discrete indicators are updated using conditional or conjugate steps, while continuous parameters are updated using HMC or NUTS. The estimation strategy in Lu and Shimizu (2025) follows this paradigm, combining shrinkage priors with modern MCMC techniques to achieve stable and efficient inference.

2 Choice of implementation

We emphasize that we do not attempt to faithfully replicate the tailored Metropolis–Hastings (TMC) algorithm proposed in Lu(25). Instead, for the Bayesian estimation component, we apply Hamiltonian Monte Carlo (HMC) to the continuous parameters in the Lu25-style model. HMC is implemented using `TensorFlow` and `TensorFlow Probability`, which provide automatic differentiation and robust, off-the-shelf implementations of gradient-based MCMC methods. Relative to TMC, HMC avoids the need for custom proposal construction and tuning, while still enabling efficient exploration of high-dimensional continuous posteriors. Importantly, this substitution does not alter the underlying Bayesian logic of Lu(25), but rather adopts a modern, general-purpose sampler that is well suited for the estimation task considered here.

3 Answers

3.1 Are there any errors or unclear parts in the paper? If there are, please point them out and explain the unclear parts in the report.

(a) The notation errors in Appendix B: all likelihood terms are expressed in the wrong forms. For example, the second term in the first equation in B.1 is presented as

$$\pi(\bar{\beta}, \bar{\xi}, \eta, r|\cdot) \propto p(\bar{\beta}, \bar{\xi}, \eta, r|q)\pi(\bar{\beta})\pi(\bar{\xi})\pi(\eta)\pi(r) \tag{1}$$

should be

$$p(q|\bar{\beta}, \bar{\xi}, \eta, r). \tag{2}$$

Another example is that, the posterior probability in page 49

$$\hat{\theta} = \arg\max_{\theta} \log[L(\theta|\cdot)\pi(\theta)] \tag{3}$$

should be

$$\hat{\theta} = \arg\max_{\theta} \log[L(\cdot|\theta)\pi(\theta)]. \tag{4}$$

In fact, all likelihood terms in Appendix B are making the same mistake as above.

(b) One point of potential confusion arises in Equation (2) in Section 2.1, where the authors define the choice probabilities $\sigma_{jt}(\cdot)$ for each good j in market t as

$$\sigma_{jt}(\xi_t, f) = \int \frac{\exp(X_{jt}^T\beta + \xi_{jt})}{1 + \sum_{k=1}^{J_t} \exp(X_{kt}^T\beta + \xi_{kt})} f(\beta)d\beta. \tag{5}$$

While this equation is mathematically correct, it hasn't clearly explained to us that the denominator implicitly includes the *outside goods* via the leading constant term 1, corresponding to $\exp(X_{0t}^T\beta + \xi_{0t})$, which is $\exp(0) = 1$.

As a result, the logarithm of likelihood in equation (3) and equation (12) shoud be unpacked as

$$\log L(\xi_{1\ldots T}, f) = \sum_{t=1}^{T}\sum_{j=1}^{J_t} q_{jt}\log\sigma_{jt}(\xi_t, f) + \sum_{t=1}^{T}(N_t - \sum_{1}^{J_t} q_{jt})\log(1-\sum_{j=1}^{J_t}\sigma_{jt}), \quad (6)$$

where both inside and outside goods are included, and N, set as 1000 in Monte Carlo simulation in section 4.1, is the number of consumers per market.

3.2 Can you replicate their results (BLP with & without cost IV, Shrinkage) as in the simulation section (Section 4)? If not, do you have any hypotheses on why?

(Shrinkage) MCMC part: We implemented with HMC and limited number of draws, as detailed in section 2. The results look good, well replicating those in Lu(25) even the number of draws is much lower than their settings.

BLP part: Across both BLP estimators, some (SD) standard deviation values of the intercept is relatively large. We believe this reflects a structural feature of the BLP estimation procedure in finite samples rather than an implementation issue: The intercept captures the overall level of mean utility and therefore absorbs common shifts in the inverted mean utilities δ_{jt} that arise from simulation noise and finite-sample variability in market shares. Such shifts tend to be approximately uniform across products within a market and thus load primarily on the intercept rather than on slope coefficients. Moreover, in the IV–GMM framework the intercept is weakly disciplined by the moment conditions, since instruments are designed to identify price endogeneity and relative demand variation rather than the absolute level of utility. As a result, even when slope coefficients are precisely estimated, the intercept can exhibit substantial sampling variability, a pattern that is common to both the "with cost IV" and "without cost IV" BLP specifications.

Table 1: Simulation results for DGP1

			BLP (with cost IV)					BLP (without cost IV)					Shrinkage					
J	T		Int	β_p	β_w	σ	ξ	Int	β_p	β_w	σ	ξ	Int	β_p	β_w	σ	ξ	Prob.
5	25	Bias	0.79	0.19	-0.29	0.06	0.18	0.46	-0.83	-0.05	0.06	0.86	1.22	-0.01	-0.00	0.01	1.22	1.00
		SD	2.25	0.50	1.59	0.43	0.36	3.42	5.35	1.57	0.50	0.87	0.39	0.11	0.02	0.10	0.39	0.23
5	100	Bias	1.11	0.05	-0.26	0.22	0.06	0.49	-1.91	0.66	0.09	0.84	0.53	0.12	-0.10	-0.21	0.71	0.83
		SD	5.58	0.52	0.79	0.65	0.99	6.21	3.68	8.08	0.60	0.20	0.38	0.37	0.10	0.26	0.27	0.08
15	25	Bias	-1.05	0.04	-0.05	0.00	0.94	-0.58	-0.69	0.06	0.02	0.71	1.22	-0.01	0.00	-0.00	1.29	1.00
		SD	6.88	0.24	0.43	0.24	0.70	1.00	6.26	1.24	0.31	0.96	0.87	0.11	0.01	0.09	0.77	0.18
15	100	Bias	0.62	0.03	-0.03	0.10	0.50	-2.89	-0.16	0.09	-0.03	0.75	0.50	0.50	-0.20	-0.50	0.70	0.90
		SD	3.63	0.25	0.27	0.51	0.30	2.67	2.91	0.97	0.41	0.39	0.00	0.00	0.00	0.00	0.00	0.10

Table 2: Simulation results for DGP2

			BLP (with cost IV)					BLP (without cost IV)					Shrinkage					
J	T		Int	β_p	β_w	σ	ξ	Int	β_p	β_w	σ	ξ	Int	β_p	β_w	σ	ξ	Prob.
5	25	Bias	1.23	0.38	-0.54	0.20	0.53	2.43	-0.09	0.21	0.09	0.68	1.07	0.02	-0.01	0.01	1.08	0.99
		SD	8.07	1.10	2.85	0.64	0.50	9.58	4.49	2.00	0.37	0.87	0.53	0.13	0.02	0.10	0.52	0.23
5	100	Bias	1.09	0.23	-0.31	0.39	0.58	-0.48	-1.71	0.47	0.05	0.72	0.58	0.04	-0.02	0.05	0.83	0.87
		SD	9.70	0.53	0.86	0.81	0.54	5.19	6.74	2.11	0.59	0.51	0.79	0.11	0.08	0.13	0.55	0.22
15	25	Bias	-0.97	0.06	-0.11	0.03	0.19	-1.77	-0.08	-0.01	-0.01	0.65	0.85	0.00	-0.00	0.01	1.22	1.00
		SD	2.83	0.29	0.47	0.43	0.04	1.16	4.22	1.13	0.29	0.81	1.25	0.11	0.01	0.09	0.88	0.18
15	100	Bias	-0.67	0.10	0.02	-0.07	0.11	-2.01	-0.22	0.07	0.01	0.10	0.50	0.47	-0.19	-0.46	0.70	0.91
		SD	1.34	0.19	0.20	0.29	0.87	4.21	4.00	1.42	0.51	0.81	0.00	0.19	0.05	0.28	0.00	0.10

Table 3: Simulation results for DGP3

			BLP (with cost IV)					BLP (without cost IV)					Shrinkage					
J	T		Int	β_p	β_w	σ	ξ	Int	β_p	β_w	σ	ξ	Int	β_p	β_w	σ	ξ	Prob.
5	25	Bias	1.54	0.02	0.19	0.09	0.58	1.00	-0.24	0.17	0.17	0.36	1.06	-0.02	-0.02	0.04	1.07	–
		SD	2.05	0.71	1.01	0.43	0.65	6.02	5.52	1.28	0.57	0.81	0.51	0.17	0.23	0.14	0.48	–
5	100	Bias	1.65	0.14	0.04	0.22	0.13	0.48	0.64	-0.02	0.06	0.73	0.55	0.49	-0.19	-0.48	0.57	–
		SD	6.19	0.23	0.93	0.57	0.20	3.63	5.78	2.23	0.45	0.54	0.37	0.05	0.04	0.10	0.37	–
15	25	Bias	-2.03	0.08	-0.11	-0.05	0.55	-0.29	-0.57	0.06	0.04	0.44	0.91	-0.02	-0.01	0.03	0.97	–
		SD	6.27	0.19	0.36	0.25	0.39	9.19	2.82	0.84	0.44	0.14	0.67	0.12	0.10	0.12	0.58	–
15	100	Bias	-3.56	0.10	-0.01	-0.10	0.55	0.05	0.17	-0.07	0.06	0.13	0.50	0.50	-0.20	-0.50	0.52	–
		SD	6.44	0.13	0.14	0.23	0.39	9.88	3.25	0.88	0.43	0.45	0.00	0.00	0.00	0.00	0.01	–

Table 4: Simulation results for DGP4

			BLP (with cost IV)					BLP (without cost IV)					Shrinkage					
J	T		Int	β_p	β_w	σ	ξ	Int	β_p	β_w	σ	ξ	Int	β_p	β_w	σ	ξ	Prob.
5	25	Bias	0.94	0.22	0.03	0.06	0.57	2.52	-0.88	0.24	0.12	0.36	1.09	0.07	-0.05	0.05	1.12	–
		SD	2.98	0.56	1.26	0.42	0.43	7.18	5.56	1.40	0.58	0.69	0.61	0.16	0.23	0.14	0.54	–
5	100	Bias	2.42	0.15	0.05	0.12	0.98	1.30	0.98	-0.25	0.07	0.29	0.57	0.41	-0.16	-0.37	0.64	–
		SD	5.16	0.25	0.70	0.50	0.53	3.10	3.43	1.18	0.44	0.66	0.46	0.18	0.08	0.22	0.39	–
15	25	Bias	-1.31	0.08	-0.14	-0.03	0.49	0.43	-0.64	0.03	0.06	0.11	0.92	0.07	-0.04	0.03	0.98	–
		SD	5.32	0.18	0.30	0.21	0.24	9.99	2.72	0.74	0.42	0.15	0.66	0.14	0.09	0.13	0.58	–
15	100	Bias	-2.97	0.09	-0.03	-0.09	0.44	0.07	0.37	-0.12	0.07	0.27	0.50	0.50	-0.20	-0.50	0.52	–
		SD	4.32	0.12	0.14	0.15	0.19	10.94	3.35	1.03	0.53	0.50	0.00	0.00	0.00	0.00	0.01	–

3.3 Explain the benchmark models used in detail. What are the assumptions needed for the BLP + IV to work in the case of price endogeneity? Delineate key variables that may or may not be observed in choice model data that justifies the IV approach, and how to find suitable instruments that satisfy the required assumptions (which you should state). Provide and justify at least three examples of viable instruments in the credit card offer setting. Can you think of another benchmark model that could be used?

3.3.1 Explanation of the Benchmark Models

The framework. This paper simulate demand estimation using two benchmark models based on the classic Berry–Levinsohn–Pakes (BLP) framework with one random coefficient on price and instrumental variables (IVs) to address price endogeneity.

Specifically, Consumers in market t choose among J inside goods and an outside option. The utility of consumer i for product j in market t is given by

$$u_{ijt} = \delta_{jt} + \sigma \nu_i p_{jt} + \varepsilon_{ijt},$$

where δ_{jt} is the mean utility, p_{jt} is price, $\nu_i \sim \mathcal{N}(0,1)$ captures heterogeneous price sensitivity, $\sigma > 0$ is the standard deviation of the random coefficient, and ε_{ijt} follows the Type I extreme value distribution. The outside option utility is normalized to zero.

Given (δ_{jt}, σ), the predicted market share for product j in market t is approximated via Monte Carlo integration:

$$\hat{s}_{jt}(\delta, \sigma) = \frac{1}{R} \sum_{r=1}^{R} \frac{\exp(\delta_{jt} + \sigma \nu_r p_{jt})}{1 + \sum_{k=1}^{J} \exp(\delta_{kt} + \sigma \nu_r p_{kt})},$$

where $\{\nu_r\}_{r=1}^{R}$ are R draws from $\mathcal{N}(0,1)$.

Berry Contraction Mapping. For any candidate value of σ, the Berry contraction mapping recovers the vector of mean utilities $\delta(\sigma) = \{\delta_{jt}(\sigma)\}$ that solves

$$\hat{s}_{jt}(\delta(\sigma), \sigma) = s_{jt}^{\text{obs}},$$

where s_{jt}^{obs} are observed market shares. This is achieved using the Berry (1994) contraction mapping:

$$\delta_{jt}^{(k+1)}(\sigma) = \delta_{jt}^{(k)}(\sigma) + \log s_{jt}^{\text{obs}} - \log \hat{s}_{jt}(\delta^{(k)}(\sigma), \sigma),$$

initialized at

$$\delta_{jt}^{(0)} = \log s_{jt}^{\text{obs}} - \log s_{0t}^{\text{obs}}, \quad s_{0t}^{\text{obs}} = 1 - \sum_{j=1}^{J} s_{jt}^{\text{obs}}.$$

The iteration continues until convergence.

Linear Decomposition and Price Endogeneity. The recovered mean utilities satisfy the linear relationship

$$\delta_{jt} = x_{jt}'\beta + \xi_{jt},$$

where $x_{jt} = (1, p_{jt}, w_{jt})$ includes an intercept, price, and observed product characteristics, and $\xi_{jt}(\sigma)$ is an unobserved demand shock. Since prices are set by firms with knowledge of $\xi_{jt}(\sigma)$, price is endogenous:

$$\mathbb{E}[p_{jt}\xi_{jt}(\sigma)] \neq 0.$$

As a result, ordinary least squares estimation of β is inconsistent.

Instrumental Variables. To address endogeneity, the model employs instrumental variables Z_{jt} that satisfy:

$$\mathbb{E}[Z_{jt}\xi_{jt}] = 0 \quad \text{(exogeneity)}, \qquad \text{rank}(\mathbb{E}[Z_{jt}x_{jt}']) = \text{full} \quad \text{(relevance)}.$$

In this paper two specific IV configurations are used:

- **BLP with Cost IV**: Instruments are

  $$Z_{jt}^{\text{cost}} = \left(1, w_{jt}, w_{jt}^2, u_{jt}, u_{jt}^2\right),$$

  where w_{jt} is an observed product characteristic (e.g., product quality or size) and u_{jt} is an exogenous cost shock that affects the price but not demand directly. This set is considered strong because u_{jt} is excluded from the demand equation and hence satisfies the exclusion restriction.

- **BLP without Cost IV**: Instruments include only functions of w_{jt}:

  $$Z_{jt}^{\text{nocost}} = \left(1, w_{jt}, w_{jt}^2, w_{jt}^3, w_{jt}^4\right).$$

  This IV set is considered weaker because all instruments are functions of w_{jt}, which may be highly collinear and do not leverage variation from cost shocks. This setup is included in the simulation to highlight problems of identification with weak IVs.

2SLS Estimation of Linear Parameters. For each fixed σ, estimate β via two-stage least squares:

$$\hat{\beta}(\sigma) = (X'P_Z X)^{-1} X' P_Z \delta(\sigma), \quad P_Z = Z(Z'Z)^{-1}Z',$$

with residuals

$$\hat{\xi}(\sigma) = \delta(\sigma) - X\hat{\beta}(\sigma).$$

GMM Estimation of σ. The nonlinear parameter σ is estimated by minimizing the GMM objective:

$$Q(\sigma) = \hat{g}(\sigma)'W\hat{g}(\sigma), \quad \hat{g}(\sigma) = \frac{1}{TJ}Z'\hat{\xi}(\sigma), \quad W = (Z'Z)^{-1}.$$

This is implemented via bounded scalar optimization over a plausible range of σ values (e.g., [0.001, 5.0]). At the minimizing $\hat{\sigma}$, the model recomputes $\hat{\delta}$, $\hat{\beta}$, and $\hat{\xi}$.

Summary. Both benchmark estimators follow the same procedure: (i) Berry contraction mapping for δ, (ii) 2SLS for β, and (iii) GMM optimization for σ. The difference lies in the choice of instruments:

- *BLP with cost IV* uses exogenous cost shocks u_{jt}.
- *BLP without cost IV* relies on higher-order polynomials of w_{jt} only.

3.3.2 Assumptions for BLP + IV with Endogenous Price

The BLP estimator assumes:

- **Price Endogeneity**: The price p_{jt} is correlated with unobserved demand shocks ξ_{jt} (e.g., firm knows ξ_{jt} and sets p_{jt} accordingly).
- **Exclusion Restriction**: Instruments Z_{jt} must be correlated with p_{jt} but uncorrelated with ξ_{jt}.
- **Instrument Relevance**: $\text{Cov}(Z_{jt}, p_{jt}) \neq 0$.
- **Rank Condition**: The instrument matrix Z has full rank (no multicollinearity).
- **Independent Utility Errors**: Consumer idiosyncratic errors ε_{ijt} follow a type I extreme value distribution.

3.3.3 Observable vs Unobservable Variables and IV Justification

In discrete choice models, the following components arise:

- **Observed:**
  - w_{jt}: Observable product characteristics (e.g., features, ratings)
  - q_{jt} or s_{jt}: Market share or quantity sold
  - p_{jt}: Price (endogenous)
- **Unobserved:**
  - ξ_{jt}: Unobserved product-market appeal (latent utility shock)
  - u_{jt}: Cost shock (unobservable in real-world data but included in DGP for simulations)

The IV approach is justified when prices are endogenously determined as a function of ξ_{jt}. Valid instruments are those that affect p_{jt} but not ξ_{jt} directly.

3.3.4 Examples of Valid Instruments in a Credit Card Offer Setting

In a real-world setting such as consumer choice over credit card offers, the following may serve as valid instruments:

(a) **Cost Shocks**: Bank-specific promotional costs (e.g., marketing budget allocated for the quarter). These influence price but are orthogonal to consumer taste shocks.

(b) **Lagged Characteristics**: Past period's interest rate or annual fee, assuming they are correlated with current pricing but not current shocks.

(c) **Other-Product Averages**: Average characteristics of competing products (excluding product j) in the same market, e.g.,

$$\bar{w}_{-j,t} = \frac{1}{J-1} \sum_{k \neq j} w_{kt}.$$

This type of BLP-style IV captures equilibrium pricing interactions but is excluded from the utility of product j.

Each of these satisfies:

- **Relevance**: Correlated with p_{jt} due to firm pricing behavior.
- **Exogeneity**: Uncorrelated with ξ_{jt}, as they are not demand shifters.

3.3.5 Alternative Benchmark Models

Another possible benchmark model could be the **Control Function Approach**. Instead of using IVs explicitly, it models the endogeneity via:

$$p_{jt} = Z_{jt}\pi + v_{jt}, \quad \text{and then include } \hat{v}_{jt} \text{ in the utility model.}$$

This approach assumes joint normality and is semiparametric, providing more flexibility when instrument strength or exclusion is uncertain.

Other options include:

- **Nested Logit**: If products can be grouped into nests (e.g., by issuer), allowing for correlated utility shocks within nests.
- **Random Forest IVs (Machine Learning IV)**: If a large number of potential instruments are available and selection needs regularization.

3.4 How can we modify the deep context-dependent choice model of Zhang(25) with the same assumption as in Lu(25)? Can you design a simulation study and implement your idea in choice-learn to show that your modification is correct?

3.4.1 Modifying DeepHalo, Zhang(25), under the Lu(25) Sparse-Shock Assumption

DeepHalo models choice behavior using a set-based neural network that maps the offered assortment and its context directly into utilities, implicitly assuming that all systematic variation in demand can be explained by the network. In contrast, Lu(25) posit that observed choice probabilities may contain *rare, additive, market–product-specific shocks* that should not be absorbed by the systematic component. These shocks are sparse across products and markets and are identified via sparsity-inducing regularization.

To reconcile these two perspectives, we modify DeepHalo by decomposing mean utility into a systematic, set-based component and a sparse additive shock term. Specifically, for choice occasion b (In our notation, b corresponds to Lu25's market–time index (i, t)) and product j (which remains as the product index), we specify

$$u_{bj} = u_j(X_b; \theta) + \xi_j + \varepsilon_{bj},$$

where $X_b \in \{0, 1\}^J$ denotes the offered set, $u_j(\cdot; \theta)$ is the DeepHalo network parameterized by θ, ξ_j is an additive product-level shock, and ε_{bj} is an i.i.d. Type-I extreme value error. This specification directly mirrors the structure of Lu(25), with the linear index $X\beta$ replaced by a flexible set-based neural representation.

Let $y_b \in \{1, \ldots, J\}$ denote the observed choice in choice occasion b, with $y_{bj} = \mathbb{I}\{y_b = j\}$ denoting the corresponding one-hot indicator. Under the assumption that ε_{bj} follows an i.i.d. Type-I extreme value distribution, the induced choice probability takes the multinomial logit form

$$P(y_b = j \mid X_b; \theta, \xi) = \frac{\exp(u_j(X_b; \theta) + \xi_j)}{\sum_{k \in X_b} \exp(u_k(X_b; \theta) + \xi_k)},$$

where the summation is taken over the offered set X_b. Substituting this expression into the likelihood yields a differentiable objective in both θ and ξ, which we optimize jointly via backpropagation using stochastic gradient descent. In practice, we employ the Adam optimizer and sweep over the sparsity penalty λ_ξ to study the bias–variance tradeoff in shock recovery.

We estimate (θ, ξ) by regularized maximum likelihood,

$$\min_{\theta,\xi} \; -\sum_b \log P(y_b \mid X_b; \theta, \xi) + \lambda_\xi \|\xi\|_1,$$

where the ℓ_1 penalty enforces sparsity in ξ and prevents the neural network from absorbing rare idiosyncratic shocks. This two-component structure preserves DeepHalo's ability to capture rich substitution patterns while explicitly accommodating Lu–Shimizu-style sparse demand shocks.

3.4.2 Simulation Design

We validate the proposed modification using a synthetic simulation study. Data are generated from a known DeepHalo *teacher* network augmented with a sparse additive shock vector. For each choice occasion, we randomly draw a variable-size assortment, compute systematic utilities using the teacher DeepHalo model, and add a sparse shock vector $\xi^\star$ whose nonzero entries are randomly selected. Choices are then sampled from the resulting softmax probabilities. We compare two estimators: (i) a baseline DeepHalo model that omits the shock term, and (ii) the modified DeepHalo+ξ model estimated under an ℓ_1 penalty.

As the result, Figure 1 reports the validation negative log-likelihood (NLL) of the modified DeepHalo+ξ model as a function of the sparsity penalty λ_ξ, together with the baseline DeepHalo model that omits the shock term. For small values of λ_ξ, allowing for an additive sparse shock component leads to a clear improvement in out-of-sample fit relative to the baseline, indicating that the DeepHalo network alone is unable to fully absorb the idiosyncratic demand shocks present in the data-generating process.

As λ_ξ increases, the penalty increasingly shrinks ξ toward zero, and the model's performance gradually converges to—and eventually becomes worse than—that of the baseline model. This pattern reflects the bias–variance tradeoff inherent in sparse shock estimation: when λ_ξ is too large, the shock component is over-regularized and the model is forced to explain all variation through the systematic DeepHalo component, reintroducing misspecification. The U-shaped relationship between validation NLL and λ_ξ therefore provides direct empirical evidence supporting the Lu(25) assumption that rare additive shocks exist and should be modeled separately from the systematic utility component.

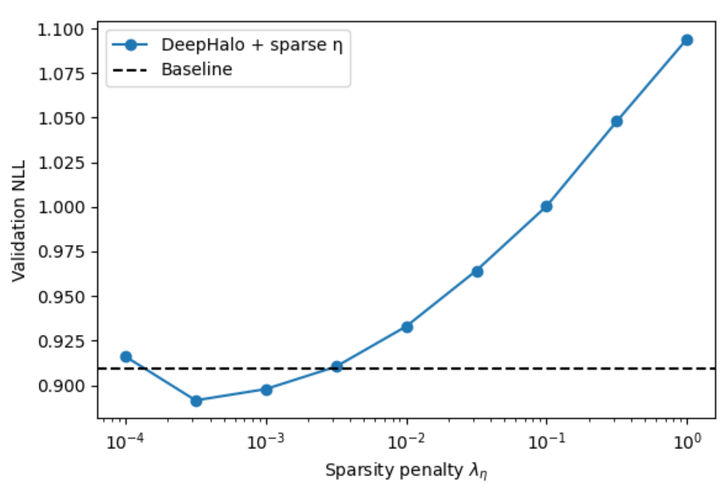

Figure 1: Validation negative log-likelihood as a function of the sparsity penalty λ_ξ. (NOTE: we shall replace the typo η as ξ in this figure but haven't done so. Please bear it in mind.)

3.5 Do you think the sparse assumption of Lu(25) applies to the credit card offer problem?

The sparse unobserved shock assumption in Lu and Shimizu (2025) is plausible for credit card offer environments in which demand responses are driven by discrete, event-based factors such as limited-time sign-up bonuses, targeted marketing campaigns, fee waivers, or regulatory or media events. In these settings, large unobserved demand shocks tend to occur only for a small subset of card–segment–time combinations, while most product–market pairs experience negligible shocks, making sparsity a reasonable and useful modeling approximation.

However, the assumption is less appropriate when demand evolves smoothly over time due to habit formation, gradual changes in consumer creditworthiness, or persistent rewards programs that apply broadly across cards and periods. In such cases, unobserved components are likely to be dense or dynamically structured rather than sparse. Overall, sparsity is best viewed as a context-dependent modeling choice that is well suited to episodic, targeted credit card promotions but may require extension or modification in environments characterized by pervasive or smoothly varying demand drivers.

4 Testing plans and testing results

To validate the correctness of our implementation, we conducted a suite of unit tests targeting structural and numerical invariants implied by the model rather than exact finite-sample equalities. For the data-generating process, tests verify that simulated market shares are non-negative, sum to less than one, and that sparsity and endogeneity patterns match the theoretical DGP definitions. For the BLP estimator, tests focus on numerical properties of the Berry contraction mapping, including deterministic reproducibility given fixed simulation draws and convergence according to the algorithm's internal stopping criterion, rather than exact reproduction of observed shares, which is not guaranteed under finite Monte Carlo integration. We additionally verify that the simulated choice probabilities reduce to the closed-form multinomial logit formula when the random-coefficient variance is set to zero. For the Bayesian shrinkage estimator, tests check prior support, latent-indicator behavior, and basic posterior sanity conditions. These tests ensure that each component of the estimation pipeline behaves consistently with its theoretical specification while acknowledging the inherent stochastic approximation error present in simulation-based demand estimation.

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